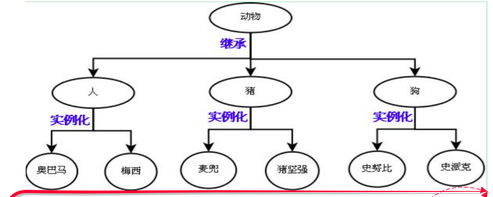
图灵python大海老师面向对象进阶

1.继承的理解



总结对象的相似之处得到了类

类实例化得到对象

总结类的相似之处得到父类

类继承父类

2.利用继承来解决冗余问题

```
class People:
    school = '图灵学院'
    def __init__(self,name,age,sex):
        self.name = name
        self.age = age
        self.sex = sex
class Student(People):
    # def __init__(self,name,age,sex):
    #     self.name = name
    #     self.age = age
    #     self.sex = sex
    def play(self):
        print('%s play football'%self.name)

class Teacher(People):
    # def __init__(self,name,age,sex):
    #     self.name = name
    #     self.age = age
    #     self.sex = sex
    def course(self):
        print('%s course'%self.name)
```

相同

```
class People:
    school = '图灵学院'
    def __init__(self,name,age,sex):
        self.name = name
        self.age = age
        self.sex = sex
class Student(People):
    # 写个继承参数
    def __init__(self,name,age,sex,score=0):
        People.__init__(self,name,age,sex)
        self.score = score
```

与继承无关

3.子类派生的新方法中重用父类功能

方式一　1.指名道姓地引用某一个类中的函数

方式二　2.super()调用父类方法

```
class People:
    school = '图灵学院'
    def __init__(self,name,age,sex):
        self.name = name
        self.age = age
        self.sex = sex

class Teacher(People):
    def __init__(self,name,age,sex,hobby):
        # People.__init__(self,name,age,sex)
        # self.hobby = hobby
        super().__init__(name,age,sex)
        self.hobby = hobby
    def course(self):
        print('%s course'%self.name)
```

和继承有关

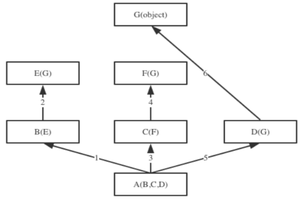
4.在单继承背景下属性的查找优先级

对象->对象的类->父类->父类.....->object

图灵大海老师

5.在多继承背景下的属性查找

广度优先查找



6.组合

概念　一个对象拥有一个属性,该属性的值是另外一个类的对象

```
class Foo:
    xxx = 100
class Bar:
    yyy = 200
    def zzz(self):
        print('我是bar方法')
obj = Foo()
obj1 = Bar()
# # obj.attr = '大海'
# # print(obj.attr)
obj.attr = obj1
```

组合单个属性　给一个对象添加一个对象属性

```
python = Course('python开发',10000)
math = Course('高等数学',3000)
English = Course('英语',4000)
music = Course('音乐',5000)
drawing = Course('绘画',6000)

# music.tell_info()
tea = Teacher('大海',24,'man')

tea.course.append(python)
tea.course.append(math)
tea.course.append(music)
```

```
        self.age = age
        self.sex = sex
class Student(People):
    def __init__(self, name, age, sex,score=0):
        super().__init__(name, age, sex)
        self.score = score
class Teacher(People):
    def __init__(self, name, age, sex,course=[]):
        super().__init__(name, age, sex)
        self.course = course

    def tell_infol(self):
        # print(self.course)
        for i in self.course:
```

组合多个属性　利用容器空列表作为属性添加多个对象

7.多态

多态性的精髓　统一

```
# 学车，学的是小汽车标准，会了所有的小汽车都会开

obj1.speak()
obj2.speak()
obj3.speak()
```

8.常用的魔法方法

__str__　打印对象直接打印__str__方法返回值

__del__　对象被删除自动调用

__call__　在对象被调用时会自动触发该方法